Hochschule
München
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SHAPEIC: DESIGN-SPACE EXPLORATION

Hierarchical LUT-based exploration from electrical models to layout-aware design

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AEMY Group

Analog Sizing: Optimize or Explore ?

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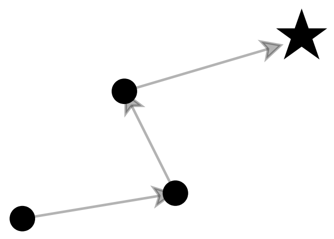
- Optimization-driven



- ✓ Targeted evaluations
- ✓ Many algorithms available
- × Partial View of the design space
- × May converge to local optima

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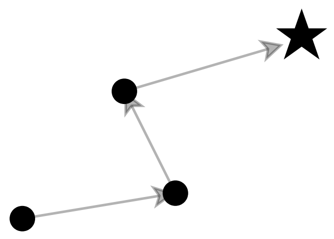

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× Partial View of the design space
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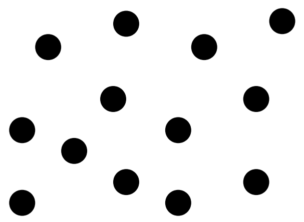
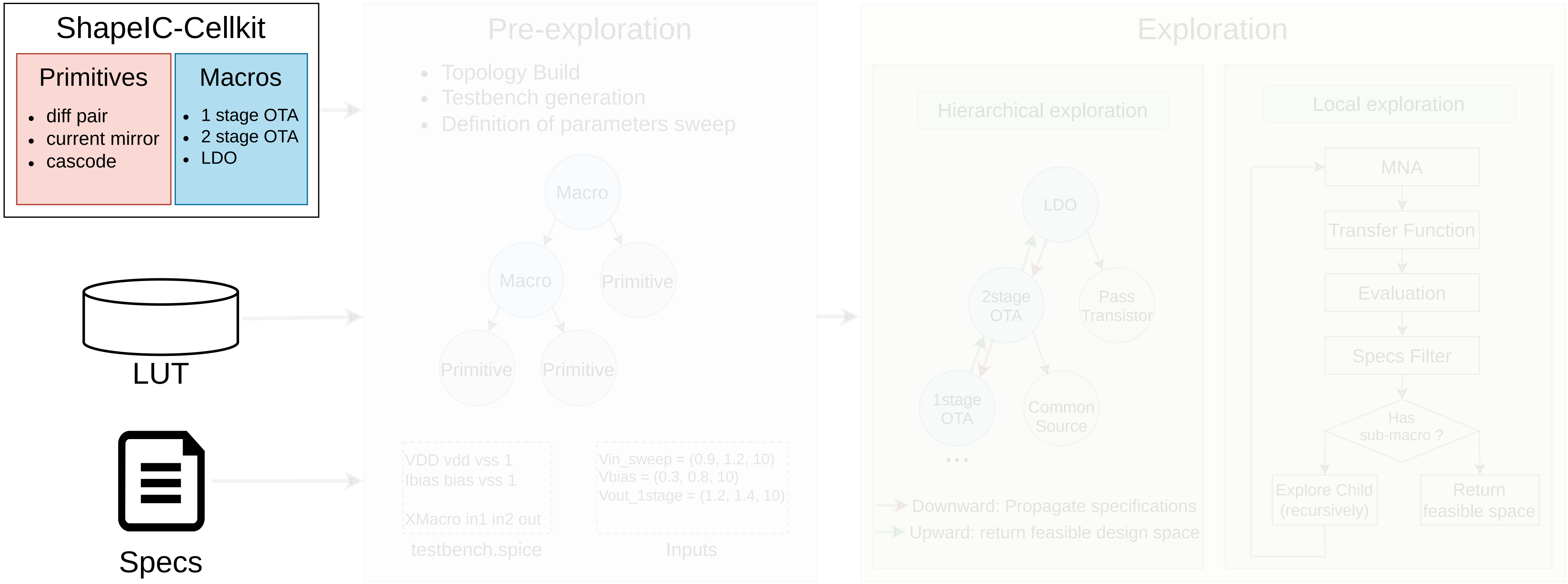
- Design-space exploration



✓ Design insight
✓ Alternative solutions
× Expensive to build
× Scales poorly without abstraction

ShapeIC: Building the Design Space



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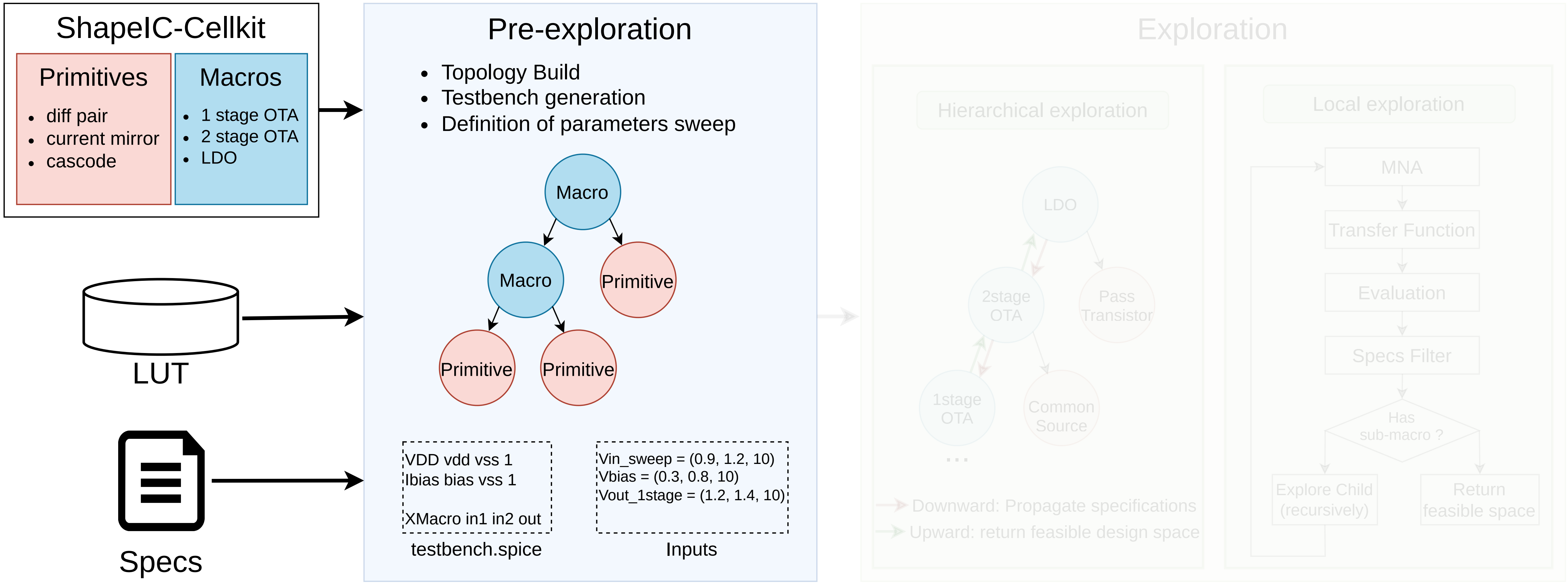

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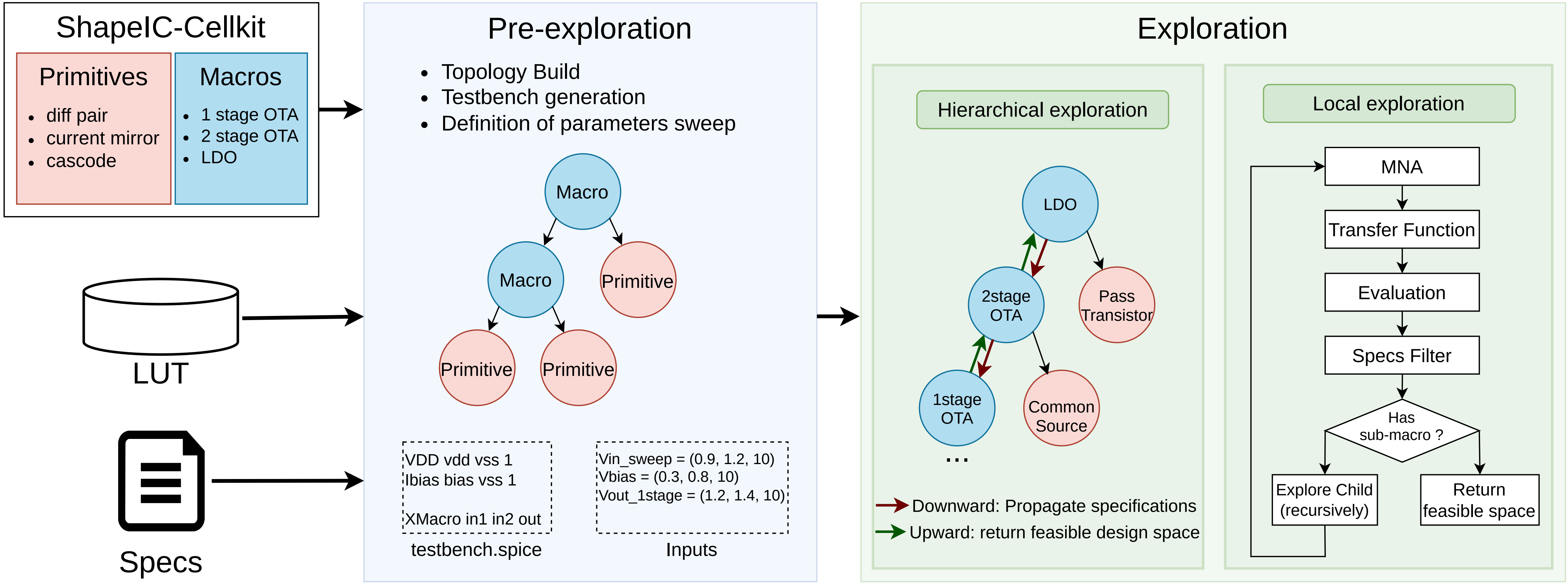
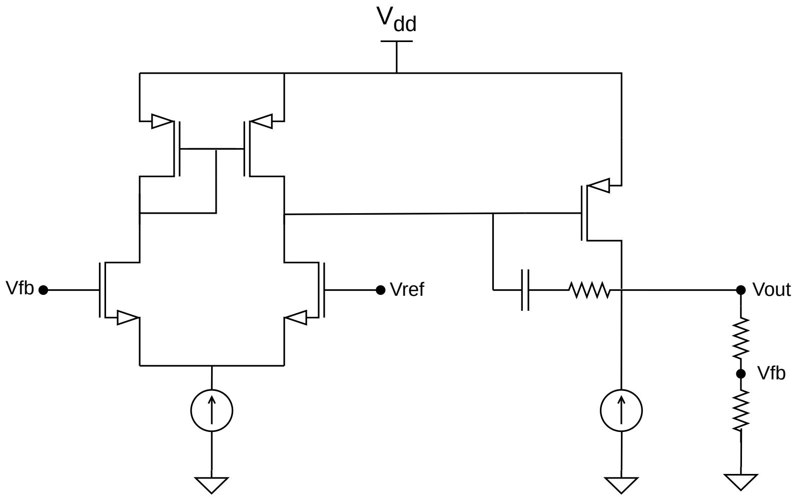


ShapeIC in Action: 2 Stage OTA



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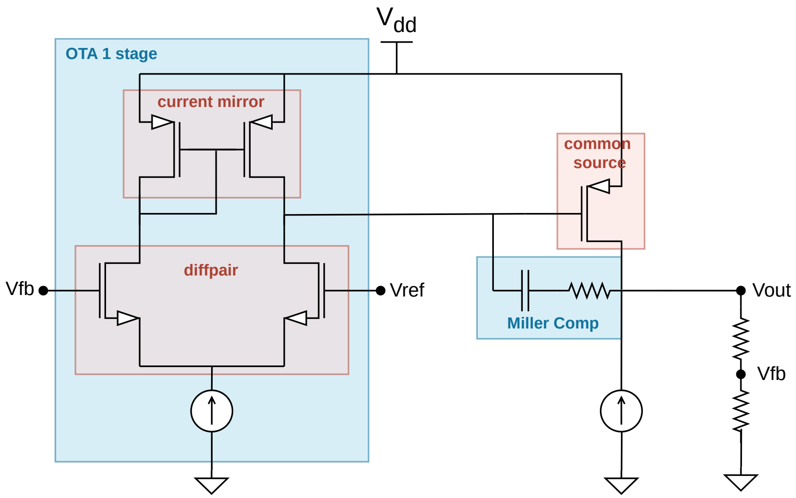

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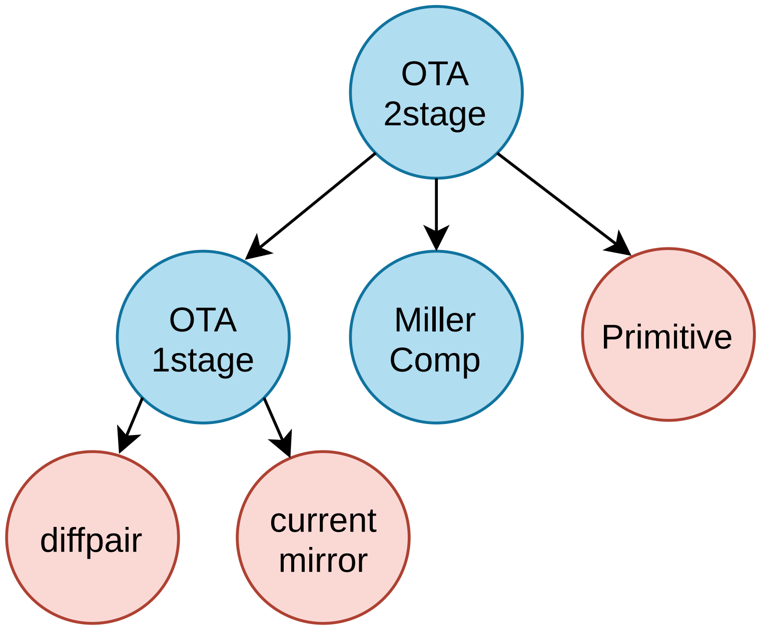
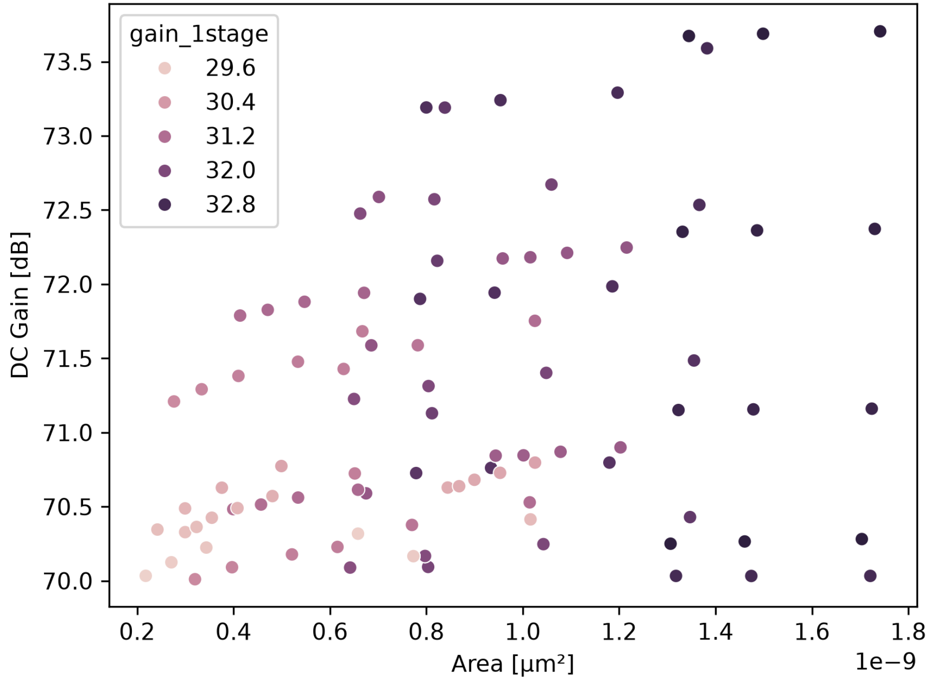


- Testbench: AC analysis
- Specs:
  $Gain \geq 70, BW \geq 10kHz, PM \geq 60$

ShapeIC in Action: 2 Stage OTA

1. Runtime
   - Exploration only: $\approx 2.79s$
   - Full run incl. LUT load: $\approx 8.38s$
2. Hierarchical exploration
   - Parent preview: $500k \rightarrow 2021$
   - Derived constraint: $Gain \geq 28.89\ [dB]$
   - Child exploration: $1050 \rightarrow 446$
   - Parent refresh: $223k \rightarrow 224$



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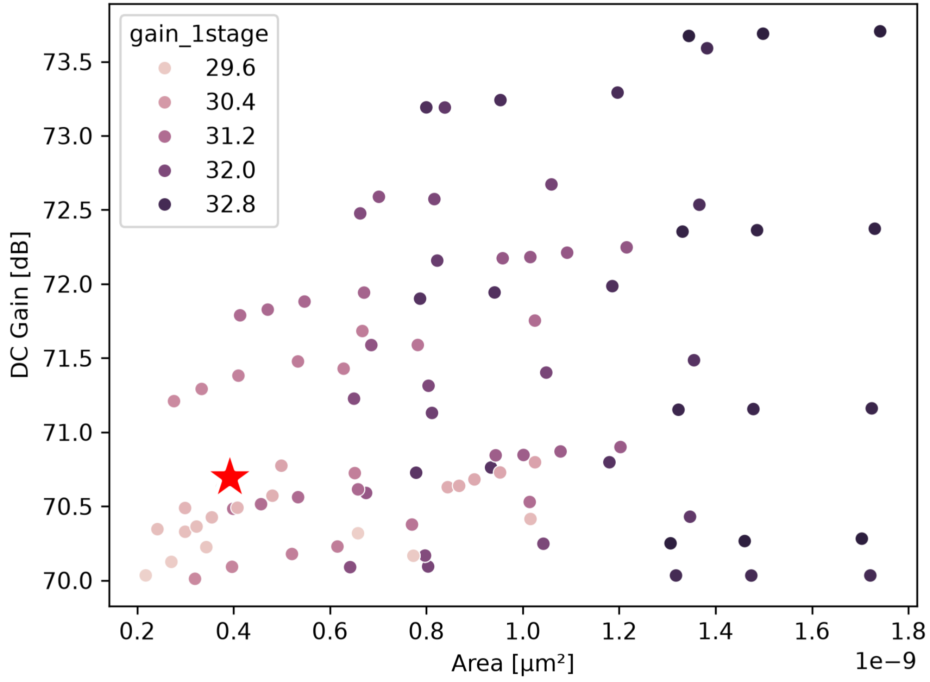
3. Simulation-based optimization
   - Random initialization: no feasible solution found
   - ShapeIC-seeded initialization: converged to a feasible solution ($\star$)



Layout-Aware Exploration

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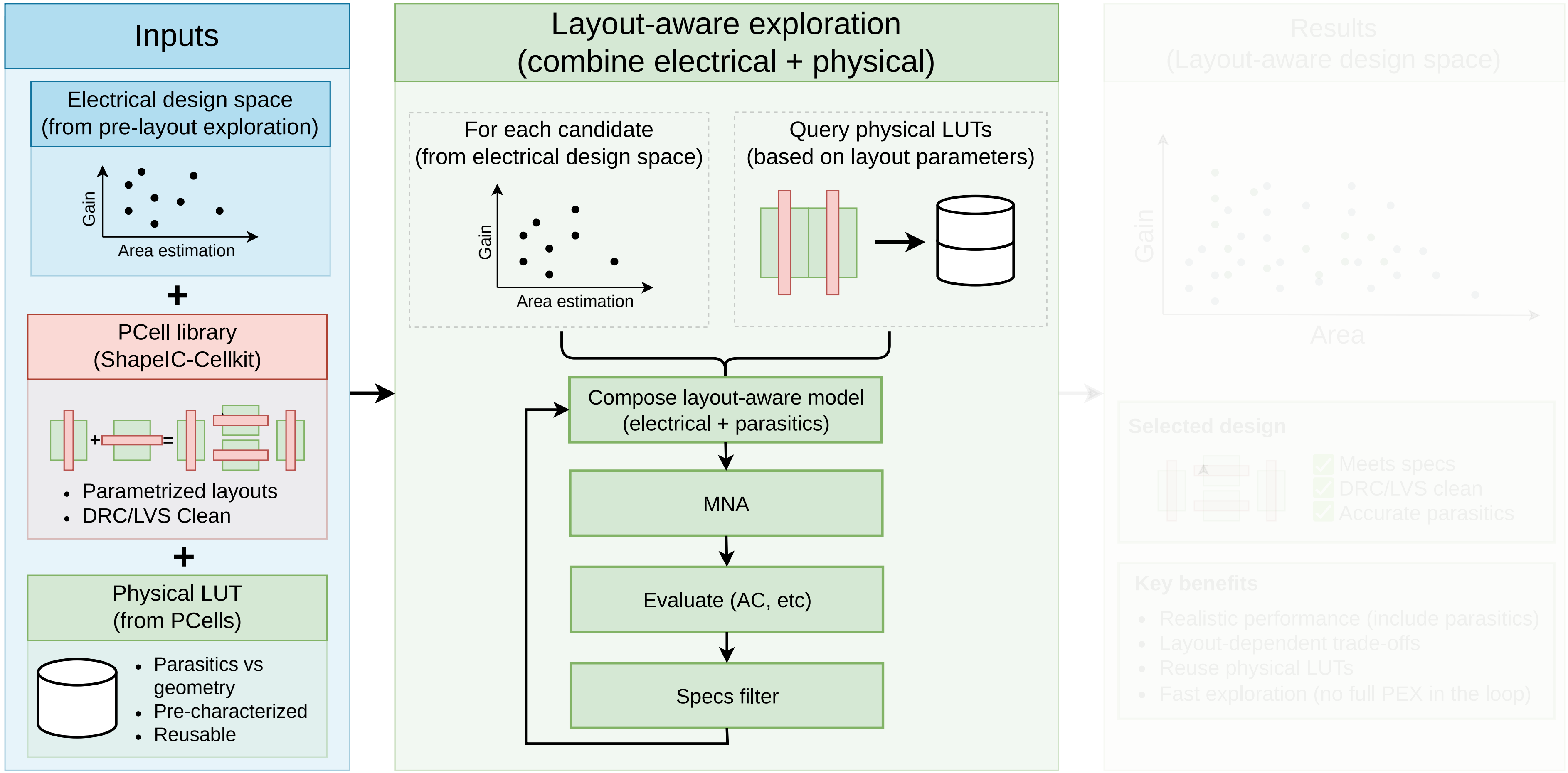

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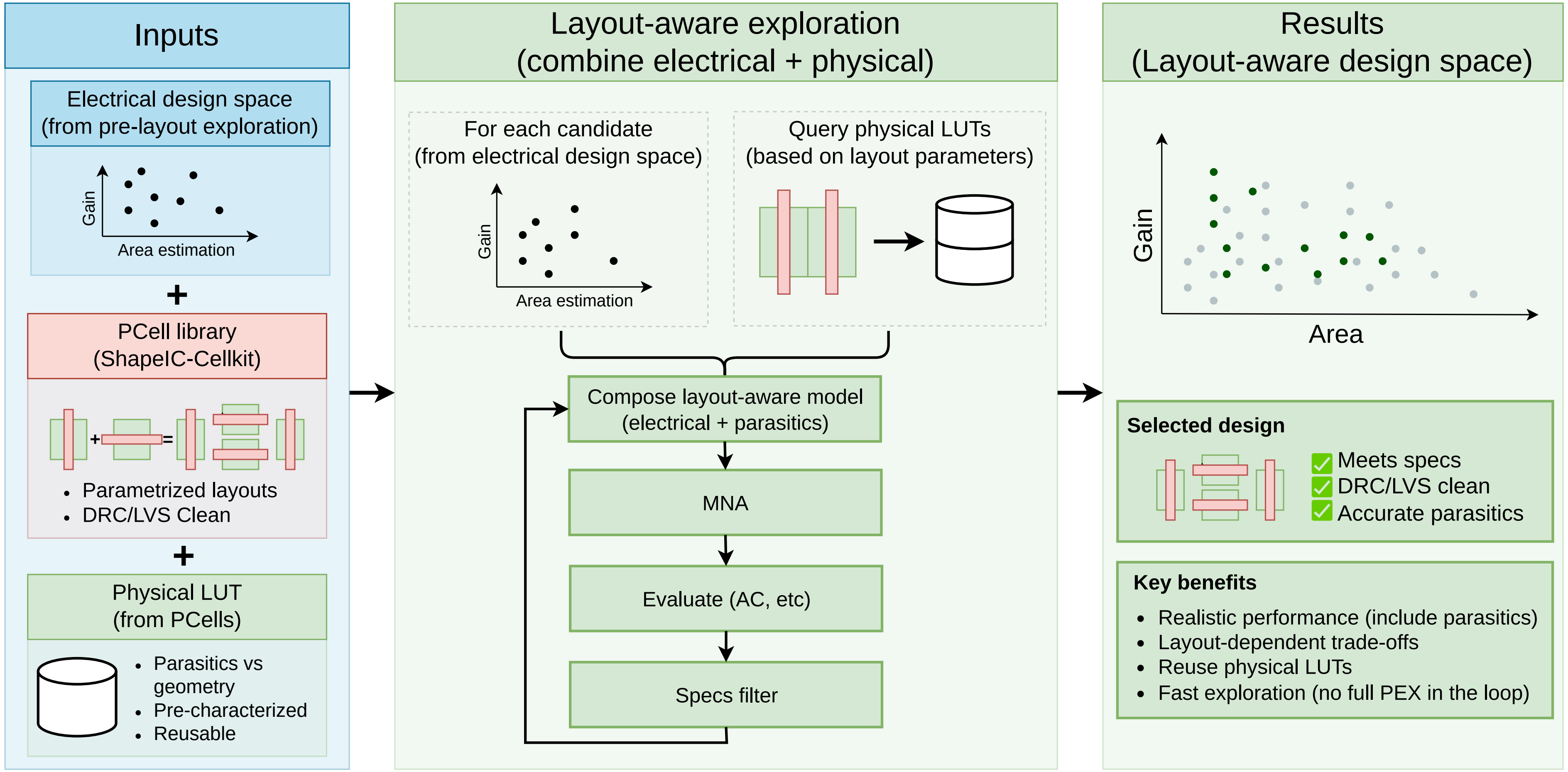

Current Status & What is Next ?

- Current status
  - AC analysis supported
  - Hierarchycal LUT-based exploration
  - Layout-aware exploration prototype
  - ShapeIC-Cellkit integration

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